



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

॥ वसुधैव कुटुम्बकम् ॥ Re-Accredited by NAAC with 'A++' Grade

Name: Dharmit Kunal Shah

PRN: 25070122073

CSE A-2

Subject: ProgrammingwithJava

ASSIGNMENT No : 4

TITLE :Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required. **Outcome**

```
public class Main {  
    public static void main (String[] args) {  
        System.out.println("Wrapped and Unwrapped variables : \n");  
  
        int x = 10;  
        Integer WrappedX = Integer.valueOf(x);  
        int unwrappedX = WrappedX.intValue();  
        System.out.println("Wrapped Int : " + WrappedX);  
        System.out.println("Unwrapped Int : " + unwrappedX + "\n");  
  
        char ch = 'A';  
        Character ch_wrapped = Character.valueOf(ch);  
        char ch_unwrapped = ch_wrapped.charValue();  
  
        System.out.println("Wrapped Char : " + ch_wrapped);  
        System.out.println("Unwrapped Char : " + ch_unwrapped);  
    }  
}
```

Wrapped and Unwrapped variables :

```
Wrapped Int : 10  
Unwrapped Int : 10  
  
Wrapped Char : A  
Unwrapped Char : A
```

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

```
public class Ex1 {  
    Run | Debug  
    public static void main (String[] args) {  
        String marks1 = "99";  
        String marks2 = "95";  
        String marks3 = "78";  
  
        Float marks_wrapped1 = Float.valueOf(marks1);  
        Float marks_wrapped2 = Float.valueOf(marks2);  
        Float marks_wrapped3 = Float.valueOf(marks3);  
  
        float total_marks = marks_wrapped1 + marks_wrapped2 + marks_wrapped3;  
        float percentage = total_marks / 3;  
  
        System.out.println("Percentage : " + percentage + " %");  
        System.out.println("Total Marks : " + total_marks + " / 300");  
    }  
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp4\Exercise> java Ex1.java  
Percentage : 90.666664 %  
Total Marks : 272.0 / 300
```

2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```
import java.util.Scanner;
```

```
class Employee {
```

```
    int id;  
    int salary;  
    int bonus;
```

```
    Employee (int i, int s, int b) {
```

```
        id = i;  
        salary = s;  
        bonus = b;  
    }
```

```
    int net_salary () {
```

```
        return salary + bonus;  
    }
```

```
    void display () {
```

```
        System.out.println("\nEmployee Details : ");  
        System.out.println("ID : " + id);  
        System.out.println("Salary : Rs." + salary);  
        System.out.println("Bonus : Rs." + bonus);  
        System.out.println("Net Salary : Rs." + net_salary());  
    }  
}
```

```

public class Ex2 {
    public static void main (String[] args) {
        try (Scanner in = new Scanner(System.in)) {
            System.out.print("Enter Employee ID : ");
            String id = in.next();
            Integer id_wrapped = Integer.valueOf(id);
            if (id_wrapped < 0) {
                System.out.println("Employee ID cannot be negative.");
                System.exit(0);
            }

            System.out.print("Enter Employee Salary : ");
            String sal = in.next();
            Integer sal_wrapped = Integer.valueOf(sal);

            if (sal_wrapped < 0) {
                System.out.println("Employee Salary cannot be negative.");
                System.exit(0);
            }

            System.out.print("Enter Employee Bonus : ");
            String b = in.next();
            Integer b_wrapped = Integer.valueOf(b);

            if (b_wrapped < 0) {
                System.out.println("Employee Bonus cannot be negative.");
                System.exit(0);
            }

            Employee e = new Employee(id_wrapped, sal_wrapped, b_wrapped);
            e.display();
        }
    }
}

```

```

PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp4\Exercise> java Ex2.java
Enter Employee ID : 89
Enter Employee Salary : 890000
Enter Employee Bonus : 7000

Employee Details :
ID      : 89
Salary  : Rs.890000
Bonus   : Rs.7000
Net Salary : Rs.897000
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp4\Exercise>

```